Deep Shearlet Network for Change Detection in SAR Images

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***Abstract*— Convolutional neural networks (CNNs) can extract shift-invariant features and have been widely applied in the change detection task. However, common CNN lacks noise robustness and needs supervised data to alleviate these problems; in this article, we propose a novel deep shearlet convolutional neural network (ShearNet) for change detection in synthetic aperture radar (SAR) images. In the network, a shearlet denoising layer (SDL) is designed to enhance the representation ability of common CNN. In SDL, feature maps are decomposed into subband coefficients by shearlet transform (ST). Due to the optimal sparse representation property and high direction sensitivity of ST, the network can capture important geometric information. Then, hard-threshold shrinkage is applied to high-frequency subbands to drop small coefficients that are most likely to be noise so that reducing the effect of noise. Finally, ShearNet is trained by introducing a noise-robust loss with noisy labels. The noisy labels are obtained by deep clustering that shows more robustness than existing preclassification methods. This fine-tuning process novelly follows the paradigm of learning from noisy labels to aside the difficulty of precisely labeling samples. Our experimental results on multiple real SAR datasets show that ShearNet can boost accuracy and have better applicability for change detection in SAR images. The source code is available at https://github.com/yizhilanmaodhh/ShearNet.**

***Index Terms*— Convolutional neural networks (CNNs), deep clustering, noise-robustness loss, shearlet transform (ST), synthetic aperture radar (SAR) image change detection.**

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I. Introduction

CHANGE detection is an important task in remotely sensed image interpretation, and it has many potential applications, such as agriculture and forestry monitoring, resource exploring, natural-disaster mapping, and urban planning and sprawl analysis [1]. The commonly used data resource includes satellite optical RGB, multispectral, hyperspectral, and synthetic aperture radar (SAR) images [2]. They usually are captured by the spaceborne or airborne imaging devices. Comparatively, optical images are usually sensitive to solar illumination and atmospheric conditions. SAR sensors use echo scattering mechanisms for imaging, so they are independent of external conditions and can monitor the land surface dynamic in all weather and all time [3]. Therefore, more attention is paid to SAR image change detection techniques. Nowadays, the Gaofen-3 satellite from China, Radarsat-2 from Canada, and TerraSAR from Germany are commonly available SAR satellites.

The goal of change detection is to detect land-cover changes from two or more images acquired at the same geographic location at different times. For this goal, traditional change detection methodologies mainly include the following techniques [4].

A. Image Registration and Noise Reduction

Image registration aims to align the pixels of two images to a coordinate system according to the geographical position [5], [6]. Since registration errors have a negative impact on the effect of change detection, some researchers devote themselves to studying change detection techniques under certain registration errors [7], [8], [9]. In the change detection community, the common case is that the data has already been registered, and the effect of registration is ignored [10], [11], [12]. Noise is another important interference factor that affects the performance of change detection. The key to noise

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reduction is to preserve structural details (such as corners and edges) in data as much as possible while denoising.

B. Difference Map Generation and Feature Extraction

The most direct way to obtain a difference map is to compare two images directly, such as the subtraction operator, the log-ratio operator, and the mean-ratio techniques [3]. Feature extraction applies mathematical and statistical techniques to transform the image to another feature space, such as frequency domain and subspace, so that making transformed features more discriminating. The typical methods include the principal component analysis (PCA) [13], the slow feature analysis (SFA) [14], and the canonical correlation analysis (CCA) [15].

C. Image Segmentation, Clustering, and Supervised Classification

The purpose of image segmentation, clustering, and supervised classification is to generate the final change detection map. This process is also a marking process, i.e., the pixels belonging to the same category are assigned the same number. In the change detection community, the thresholding-based segmentation method is commonly used, such as Otsu [16] and GKI [17]. Clustering methods distinguish different regions by the similarity between pixels. The schemas with more similarity are divided into a cluster. Otherwise, they are divided into a different one. K-means and fuzzy c-means (FCM) algorithms are commonly used to measure the similarity between the extracted features and cluster them into changed and unchanged classes [3]. Different from unsupervised segmentation and clustering algorithms, supervised classifiers make use of labeled prior knowledge and, thus, present more reliable when modeling the nonlinear relationships of changed and unchanged pixels. The data to be processed can be classified by various classifiers, such as the support vector machine (SVM) [18], [19], the K-nearest-neighbor (K-NN) classifier [20], the random forest [21], the extreme learning machine (ELM) [22], [23], and the sparse representation classifier (SRC) [24].

In actual applications, the above technologies may be integrated, mixed, or iterated to achieve a high-accuracy detection result. Since the SAR imaging system is coherent, the resulting speckle noise severely contaminates its images. Therefore, speckle suppression is always the central topic of SAR image change detection. Feature extraction is an effective way to suppress noise.

With the development of algorithm design, computation power with high performance, and big data, convolutional neural networks (CNNs) achieve remarkable progress in computer vision as a supervised classifier. It naturally integrates abstract feature extraction and classification in a learnable manner. Recent studies show that CNNs have swept various applications, such as image classification [25], [26], [27], object detection or tracking [28], [29], [30], semantic segmentation [31], and human pose estimation [32]. To construct a supervised fine-tuning process with CNNs, ground truths must be provided. Note that change detection is a pixel-level task, and each pixel needs to be labeled with its category. To avoid the difficulty of manually labeling every pixel, a preclassification strategy is commonly used to obtain training samples, such as the superpixel-based method [33] and hierarchical FCM (HFCM) [34]. Since training samples selected from preclassification are used as prior label information, their quality plays an important role in the performance of the final trained network. However, existing preclassification methods are performed in shallow features, which inherits the limitations of traditional methods. Therefore, it is not always satisfactory to provide a reliable preclassification result, especially for complex ground objects with serious pollution of noise. To improve the robustness of preclassification, we cluster deep features obtained from untrained CNN to generate preclassification. The idea is inspired by deep clustering [35], where untrained CNN still shows the strong ability of feature extraction. In a sense, it can be viewed as a fixed feature extractor, such as the Gabor transform, and is more robust since it can benefit from deep hierarchy features.

To achieve better performance, the CNN-based change detection method depends not only on a robust preclassification but also on a better convolutional network structure [10], [11]. A common CNN structure consists of convolution, pooling, and fully connected layers. Through the hierarchical structure stacked by these bricks, high-level and abstract features can be obtained and be equipped with the invariance of translation, scaling, shift, and other forms of distortions [36]. However, since there is no explicit noise-robustness mechanism in common CNN, these properties are limited, leading to insufficient robustness to noise pollution in input data. As mentioned in [37], small input changes would lead to drastic output changes in the well-trained convolutional network. In particular, the speckle noise in SAR images could be transferred into the follow-up feature maps and degrade the final output prediction of CNN. To enhance the noise robustness of CNN and improve the prediction accuracy, we design the shearlet denoising layer (SDL) based on the shearlet transform (ST) and the threshold denoising algorithm, and insert the designed SDL into CNN as an explicit denoising layer. In SAR images, random speckle noise is submerged in effective signals, and it is difficult to separate random noise from effective signals in the original image domain. ST is an effective time–frequency analysis method. In the ST domain, a valid signal is of coefficient with a large amplitude, and noise interference is of a small one. Therefore, it can effectively remove noise by discarding small coefficients with the threshold method in the shearlet domain.

Based on the above analysis, we propose a novel deep shearlet convolutional neural network (ShearNet) for change detection in SAR images. In the network, an SDL is introduced into common CNN. SDL first transforms the feature maps into the frequency domain with ST and then applies the threshold method to the resulting shearlet coefficients to suppress noise and enhance the extraction ability of CNN for important structure features. Based on ShearNet, deep clustering is used to obtain a reliable preclassification result before training. Since the preclassification is an inaccuracy, we novelly treat it as noisy labels and introduce noise-robustness loss, $\mathcal{L}_{\mathrm{DMI}}$,

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to train the ShearNet. As a result, more accurate results can be obtained from the well-trained ShearNet, which is robust to noise on input and label.

1) We proposed a general SDL, which is designed in a plug-and-play way and is easily embedded into end-to-end deep networks for other tasks.
2) We proposed ShearNet by integrating SDL into common CNN to increase its noise robustness for SAR image change detection. Due to the optimal sparse representation and highly directional sensitivity of ST, ShearNet can capture more geometric information while denoising and, thus, improve the availability of detecting change for intricacies of SAR data.
3) To alleviate the difficulty of accurately labeling samples, we introduce deep clustering to generate preclassification. The imprecise preclassification results are then used as noisy labels to train ShearNet with noise-robustness loss. Therefore, our method is fully unsupervised.

The remainder of this article is organized as follows. Section II elaborates on the problem statement and the motivation of our work. Section III describes the proposed ShearNet in detail. Section IV shows how to optimize the ShearNet to be more accurate for identifying changed regions. Extensive experimental results are presented in Section V. Finally, Section VI summarizes our work.

II. Problem and Motivation

A. Problem Statement

The image change detection is to identify the changed areas from two coregistered images that are captured at different times t_1 and t_2 but in the same area. Given two coregistered images I_1 and I_2, the change detection problem can be formulated as [9], [10], [11]

$$\Omega = G(I_1, I_2) = f_D(f_1(I_1) \ominus f_2(I_2)) \tag{1}$$

where f_1 and f_2 denote feature extraction and $\ominus$ represents the difference analysis. f_D is the classification analysis that labels each pixel as change and unchange, and generates the binary change map $\Omega = (\Omega_c, \Omega_u)$, where Ω_c and Ω_u represent the sets of changed class and unchanged class, respectively. Ω has the same spatial size as I_1 and I_2. $G = \{f_1, f_2, \ominus, f_D\}$ can be the techniques mentioned above over feature extraction, difference map generation and classification, and so on.

B. Motivation of the Proposed ShearNet

1) Convolutional Neural Networks: CNN is the most widely used deep neural network model [12], [38]. When processing pixel-level change detection in SAR images with CNN, it usually is solved as a classification task [34], [39]. Given n input patches from an image I, they are represented as $X = \{x_0, x_1, \ldots, x_{n-1}\}$. Each patch with the size of $w \times w$ is generated by extracting the neighborhood centered on a pixel. The corresponding output of these input patches is $Y = \{y_0, y_1, \ldots, y_{n-1}\}$. The convolution layer uses convolution kernels with trainable parameters to perform filtering as

$$C_i^r = \rho\left(\sum_m x_k^{r-1} * K_i^r + b_i\right) \tag{2}$$

where $*$ represents a 2-D convolution, K_i^r is the ith filter in the rth convolution layer, b_i is the corresponding bias, and $\rho(\cdot)$ is a nonlinearity activation function. Pooling operation is usually following the convolution layer to downsample the output of the convolution layer. Maximum pooling and average pooling are two commonly used pooling layers. In the rth layer, the output of maximum pooling with input x_m is expressed as

$$P_i^r = \max\{x_m \downarrow p\} \tag{3}$$

where p determines the support scale of pooling.

Following a series of convolution layers and pooling layers, the concatenated hierarchical structure is ended with fully connected layers to produce the classification output [40]. In the common CNN, the convolution operation locally connects the neighborhoods of each input by sharing parameters, which reduces the number of parameters. The pooling operation is a summary statistic of input that aims to reduce the complexity while simplifying information. They both lead to translation invariance in the space domain to some extent [32], [41]. However, due to the absence of targeted denoising mechanisms, the common CNN is associated with weak noise robustness [42]. Moreover, commonly used pooling operation, such as maximum pooling, has a limited ability to protect the texture structure of the CNN feature.

To improve the issues related to the weak noise robustness and the shortage of downsampling operations, many methods based on wavelet analysis are developed. Duan et al. [36] proposed convolutional-wavelet neural networks (CWNNs) that plug the dual-tree complex wavelet transform (DT-CWT) into commonly used CNN. In the method, DT-CWT is used to decompose a feature map into eight subbands, where the average of two low-frequency subbands is designed to act as the output of the pooling layer. Fujieda et al. [41] proposed wavelet pooling that uses a second-level wavelet decomposition to replace traditional pooling operations, where the second-level high-frequency subbands are directly discarded, and the features are reconstructed by low-frequency one. In [37], wavelet-integrated CNNs (WaveCNets) use discrete wavelet transform (DWT) to decompose feature maps into low- and high-frequency components. Analogously, to enhance the noise robustness of CNN, the high-frequency components are discarded, and only the low-frequency one is used to replace the downsampling operator.

These works produce encouraging progress; however, they all simply ignore high-frequency components. Although random noise is mostly contained in high-frequency components [37], it also contains useful texture details. It may result in the loss of useful details for completely discarding high-frequency components. In addition, the frequently used wavelet transform is suitable for 1-D signals and can handle the point singularity problem almost perfectly, but its excellent characteristics are not suitable for 2-D or higher dimensional data and their insensitivity to direction causes the limitation of signal representation. Focusing on image change detection in this work, we proposed the ShearNet, which integrates ST and the threshold denoising method as a denoising layer into common CNN to capture important texture structure features and suppress unimportant noise. Compared to directly

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discard the high-frequency components, the employment of the threshold denoising method can remove noise more precisely, and meanwhile, the useful details are retained.

2) Analysis of Shearlet Representation: As a new time–frequency analysis method, ST has multiscale and multidirectional properties, good localization characteristics, and optimal sparse representation property [43]. It is suitable for data processing, such as SAR data, which contains rich directional information, such as texture, contour, and edge, and has great advantages in denoising. Due to the higher directional sensitivity for various shapes, ST has a unique capacity to grasp the intrinsic geometrical features of multidimensional data. As a consequence, the ShearNet that introduces ST and combines with the threshold denoising algorithm promises to acquire more sparse and discriminative deep features. Thus, the separability of changed parts can be enhanced by benefiting from the classic frequency-domain denoising theory for the forward inference of CNN.

The performance of the frequency-domain denoising method depends on the sparsity of the transform method. The investigation by neurophysiologists shows that the reception field of the mammalian visual cortex has spatially localized, oriented, and bandpass characteristics, which allow only a few neurons to be activated to capture important visual information [44]. For the statistical model of natural images, a representation system that maximizes sparseness can account for these properties of the receptive fields. This means that the sparsity of a representation system reflects the valid key information of vision data. The basic function of ST is comparable to these three properties. Mathematically, the property of the sparse representation can be quantitatively measured through nonlinear approximation [45].

Consider that $\mathcal{F} = \{\phi_\mu : \mu \in I\}$ denotes a basis or a tight frame for $L^2(\mathbb{R}^2)$, and then, a function f can be represented (nonlinearly) as

$$f_N = \sum_{\mu \in I_N} \langle f, \phi_\mu \rangle \phi_\mu, N \to \infty \tag{4}$$

where I_N is the set of indexes of the N-largest inner products $|\langle f, \phi_\mu \rangle|$. The best N-term approximation error is

$$\varepsilon_N = \|f - f_N\|_2^2 = \sum_{\mu \notin I_N} \left|\langle f, \phi_\mu \rangle\right|_2^2, \quad N \to \infty. \tag{5}$$

The goal of the representation system $\mathcal{F}$ is to improve the quality of the approximated function f_N and minimize the asymptotic approximation error.

For the image f that is C^2 except for discontinuities along C^2 curves, the approximation f_N obtained by the best N-term shearlet coefficients satisfies

$$\varepsilon_N \leq C(\log N)^3 N^{-2}, \quad N \to \infty. \tag{6}$$

This is very close to the theoretical optimal approximation rate (N^{-2}) [46] and is far superior to that of Fourier ($N^{-(1/2)}$) and wavelet (N^{-1}). Hence, ST provides optimally sparse approximations for images governed by edge-like structures. This means that it is able to better capture key structural features in images.

Moreover, in many multiscale geometric analysis methods, the contourlet transform also is close to the optimal approximation rate, but it achieves directional filtering by employing a directional filter bank [47]. ST has a more flexible direction filtering by shearing. This leads to the important advantage of ST over contourlet one, i.e., there are no constraints on the number of orientations for the shearing in ST [48]. Therefore, we design the denoising layer based on ST, namely SDL, leading to the ShearNet. We concretely use α-shearlet [43], [49], which introduces a scaling parameter α_j as

$$A_{\alpha_j, j} = \begin{bmatrix} 2^{2j} & 0 \\ 0 & 2^{\alpha_j j} \end{bmatrix} \tag{7}$$

where $\alpha \in (0, 2)$, which can be varied for each scale j. The introduce of the scaling parameter increases the flexibility of parabolic scaling and allows encoding anisotropic structures in an optimal way. The specific implementation algorithm can be found in [50].

III. Deep Shearlet Network (ShearNet)

For SAR image change detection, it is crucial to retain the structure information against the challenges of effectively suppressing speckle noise. To alleviate the challenges, we proposed a deep ShearNet for change detection in SAR images. An overview of the proposed ShearNet is illustrated in Fig. 1. In this section, we first elaborate on the proposed SDL. Then, the architecture of ShearNet for change detection is presented.

A. Shearlet Denoising Layer

Due to the excellent characteristics of ST, it not only is widely applied in image denoising [48], [51] but also in image inpainting [43], edge analysis and identification [52], medical image fusion [53], and so on. In this work, ST combined with the threshold method is proposed as a module of CNN to make deep features in the shearlet domain instead of the usual spatial domain. The transform-domain method is more sensitive to the mutation part of the image. Thus, the introduction of ST can enhance the ability of CNN to capture the geometry of images and boost the information extraction for anisotropic and sparse features in an interpretable way. The transform-domain method usually decomposes an image into high-frequency components and low-frequency components. Consider that noise is generally contained in the high-frequency component; existing methods keep only the low-frequency information and fully remove the high-frequency information. This would result in the loss of useful details contained in high frequencies. To better balance between the two goals of removing noise and keeping details, it is necessary to combine ST with the threshold denoising method to accurately remove noise while effectively preserving geometric features.

In the traditional shearlet-based denoising method, according to the multiscale and multidirection characteristics of ST, a series of shearlet coefficients of different scales and directions can be obtained from noisy data with ST. When the direction of the shearlet basis function is roughly the same as the signal direction, the larger shearlet coefficient

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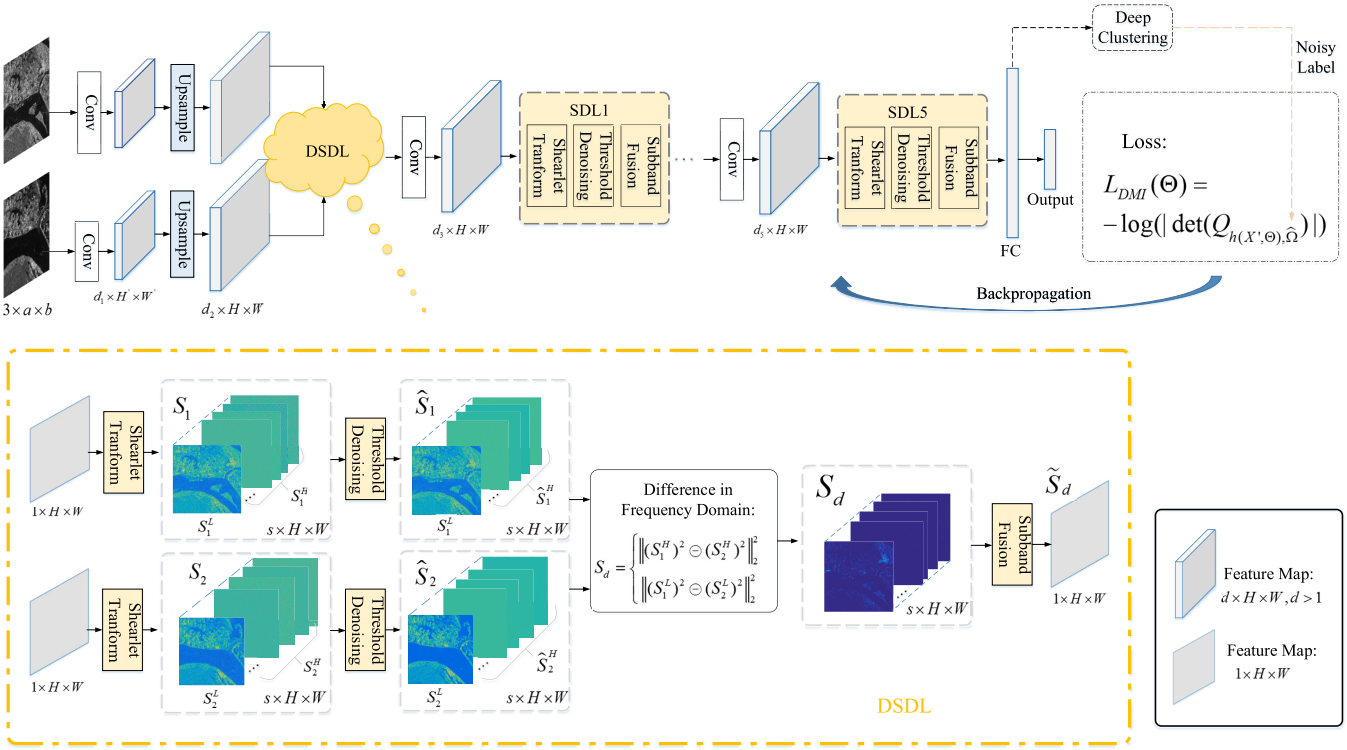


Fig. 1. Overview of the proposed ShearNet.

TABLE I

STATISTICAL ANALYSIS OF SHEARLET COEFFICIENTS. (.,.,.) REPRESENTS THE SCALE, SHEAR, AND CONE. CONE = H AND CONE = V ARE HORIZONTAL AND VERTICAL CONES

Index	Maximum	Minimum	Mean	Variance
Low Pass	168.51	8.40	45.89	32.04
(0,-1,h)	40.78	-33.03	0.06	3.57
(0,0,h)	68.44	-57.26	0.11	6.55
(0,1,h)	42.85	-28.13	0.10	3.51
(0,1,v)	28.38	-25.83	-0.02	2.18
(0,0,v)	42.51	-35.43	-0.08	5.33
(0,-1,v)	31.46	-31.50	-0.01	2.96
(1,-2,h)	32.69	-32.12	0.02	2.93
(1,-1,h)	42.81	-40.04	0.04	3.72
(1,0,h)	65.06	-61.69	0.10	5.46
(1,1,h)	56.20	-43.70	0.10	4.13
(1,2,h)	30.68	-35.09	0.0011	2.81
(1,2,v)	27.25	-24.01	0.0028	1.64

can be obtained. On the contrary, when the direction of the shearlet basis function deviates greatly from the direction of the signal, the shearlet coefficient is smaller. Therefore, the valid signal is mapped to the shearlet coefficient in a specific direction usually with larger amplitudes. Moreover, noise is not of direction, and the obtained shearlet coefficient has a small amplitude. Hereby, the threshold function is applied to remove the smaller shearlet coefficient and retain the larger part so that suppressing noise. Table I lists the statistical analysis of shearlet coefficients. It can be observed that the statistics of high-frequency subbands are significantly smaller than the low-frequency subbands.

Accordingly, the design of SDL can be summarized into three main steps. First, each CNN feature map is decomposed

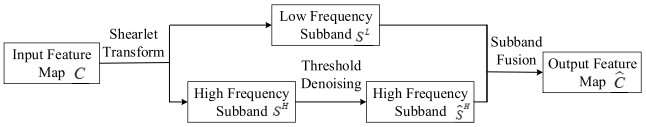


Fig. 2. SDL.

by ST. Shearlet coefficients in multiresolutions and multidirections are obtained. Then, we calculate the threshold value and apply the thresholding function to get thresholded shearlet coefficients. Finally, the denoised shearlet coefficients are aggregated to produce the fused features. The SDL framework is shown in Fig. 2. The feature map C is decomposed into different subbands of frequency and orientation. The scale parameter j, the shearing parameter k, and the cone parameter c together determine a shearlet filter in the spatial domain $\psi_{j,k,c}$. In the frequency domain, it is represented as $\hat{\psi}_{j,k,c}$. For the scale parameter $j \in \mathbb{N} \cup \{0\}$, $j = 0$ represents the low-frequency shearlet filter. In our case, we have 1 scale; j only is 0. The shearing parameter $k \in \mathbb{Z}$ ranges from $-\lceil 2^{j(1-\alpha)} \rceil$ to $\lceil 2^{j(1-\alpha)} \rceil$ on scale j for each cone. As shown in Fig. 3, the shearing level parameter partitions the corresponding ring-like passband for each scale in different directions. The frequency support of a shearlet filter is a pair of trapezoids with the approximate size of $2^{2j} \times 2^{j}$. The cone parameter $c \in \{0, 1, 2\}$ encodes the cone to which a shearlet filter belongs in the frequency domain. The frequency plane is divided into a horizontal (h: indexed by $c = 1$) cone and a vertical (v: indexed by $c = 2$) cone, and $c = 0$ indexes the low-frequency shearlet filter. In Fig. 3, the dashed part represents the vertical cone, and the solid line represents the horizontal cone. For the

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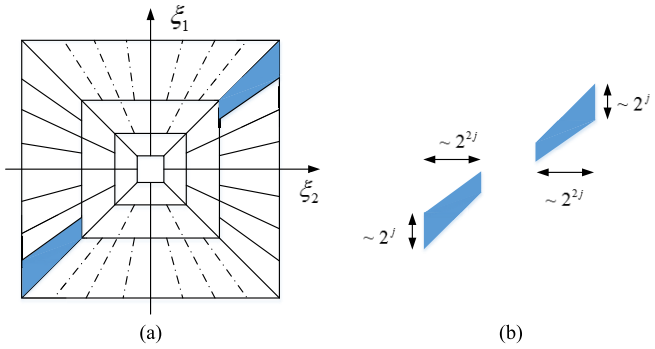


Fig. 3. Frequency support of shearlet. (a) Tilling of the frequency plane. (b) Size of the frequency support of a shearlet $\phi_{j,l,k}$.

shearlet system with $\alpha = 0.5$ and 1 scale (i.e., $j = 0$ and $k = \{-1, 0, 1\}$), each shearlet index for the shearlet parameters (j, k, c) from 1 to 7 is (0, 0, 0) (i.e., low pass component), (0, −1, h), (0, 0, h), (0, 1, h), (0, 1, v), (0, 0, v), and (0, −1, v).

In terms of thresholding operation, the key issue is the selection of the proper threshold value and the thresholding function to keep important information and kill nonimportant noise information [54]. Donoho and Johnstone [55] proposed VisuShink, which is an innovative nonlinear denoising scheme and has been widely used. Moreover, hard thresholding is a reasonable selection for SDL, since it cannot corrupt the feature distribution of hidden layer [56]. Concretely, for each of the output feature map $C_i \in \mathbb{R}^{H\times W}$ with the spatial size $H \times W$, where $i = \{0, 1, \ldots, d\}$ is the index of channel number, the one-level ST is applied. The procedure for computing the ST mainly consists of two steps, i.e., Laplacian pyramid and shearing filters [57]. The former decomposes the feature map into high- and low-frequency components at scale $j = 0$. The latter achieves the multidirection decomposition as the parameters mentioned above vary. Then, we obtain the subband coefficients with different frequencies and directions as $S_i = SH(C_i) = [S_i^L, S_i^H] \in \mathbb{R}^{s\times H\times W}$, where s denotes the number of the produced shearlet coefficients, $S_i^L \in \mathbb{R}^{1\times H\times W}$ denotes the low-frequency subband coefficient, and $S_i^H \in \mathbb{R}^{(s-1)\times H\times W}$ denotes the high-frequency subband coefficients. As mentioned above, noise is not sparse and has smaller shearlet coefficients. To suppress unimportant noise information, the universal threshold is computed as [58]

$$T_{(i,u)} = \sigma_{(i,u)}\sqrt{2\ln(N)}, \quad u = \{1, 2, \ldots, (s-1)\} \tag{8}$$

where $N = H \times W$ denotes the spatial size of the feature map and σ is noise variance and u is the index of the high-frequency subbands S_i^H. The value of the threshold guarantees the noise corresponding to a small coefficient is removed with high probability. As a result, more important information can be captured by the network. Usually, the noise variance is obtained by a robust median-based estimator as

$$\sigma_{(i,u)} = \frac{f_{\text{median}}\left(\left|S_{(i,u)}^H\right|\right)}{0.6745}, \quad u = \{1, 2, \ldots, (s-1)\} \tag{9}$$

where f_{dedian} is the median function, which reflects the characteristics of the feature itself.

Based on the calculated threshold value T, the hard threshold denoising function is applied to remove the small coefficients in the high-frequency subband S_i^H. The expression is described as follows:

$$\hat{S}_{(i,u)}^H = \begin{cases} S_{(i,u)}^H, & |S_{(i,u)}^H| \geq T_{(i,u)} \\ 0, & |S_{(i,u)}^H| < T_{(i,u)}. \end{cases} \tag{10}$$

The coefficient less than the threshold value T becomes zero, and the coefficient greater than or equal to the threshold value T remains unchanged. This threshold method has the characteristic of amplitude preservation. Thus, it does not corrupt the distribution of the feature so that enabling the network to be optimized as usual. As smaller coefficients are set to zero, this not only removes noise but also increases the network sparsity. Therefore, the network can better grasp the key features as human visual systems.

Different subbands include information on different frequencies and directions. As shown in Fig. 4, the low-frequency subband mainly encodes the structural information of the image, and high-frequency subbands encode anisotropic features in different directions. The first two rows in Fig. 4 are the original shearlet coefficients, which have an uneven surface. By contrast, the last two rows in Fig. 4 are the denoised coefficients that are more clean and smooth in the visual effects than the original high-frequency coefficients. Moreover, the different texture details are highlighted as the direction varies. For making the best use of this information, the new high-frequency subbands $\hat{S}_i^H$ and the low-frequency subband S_i^L are fused as the output of SDL

$$\hat{C}_i = S_i^L + \sum_{u=1}^{s-1} \hat{S}_{(i,u)}^H, \quad u \in \{1, 2, \ldots, s-1\} \tag{11}$$

where $\hat{C}_i \in \mathbb{R}^{H\times W}$ is the output of SDL. Accordingly, the introduce of SDL cannot only reduce noise but also enhance the direction representation of CNN. Meanwhile, it also increases the interpretability of CNN to some extent.

B. ShearNet for Change Detection

The key of our ShearNet architecture is to introduce the SDL into CNN to solve the problem of change detection. The introduce of SDL enables the network to remove noise and grasp important geometric texture information. In SDL, the highly directional sensitivity and optimal sparse representation of ST contribute to the capture of cartoon-like edge structures, and the thresholding method contributes to removing nonsignificant information and further increasing the sparsity of the network.

Fig. 1 illustrates the proposed network architecture for change detection, which is derived from AlexNet. The networks are expressed as *{Input, Conv1, Upsample, DSDL, Conv2, SDL1, . . . , Conv5, SDL5, FC, Output}*. Each convolutional block (*Conv*) contains convolution and rectified linear unit (*ReLU*) operations without pooling as it is easy to lose details. This model receives two images $I_1 \in \mathbb{R}^{3\times a\times b}$ and $I_2 \in \mathbb{R}^{3\times a\times b}$ as input, where a and b are the height and the width of the two images, respectively. Three channels are

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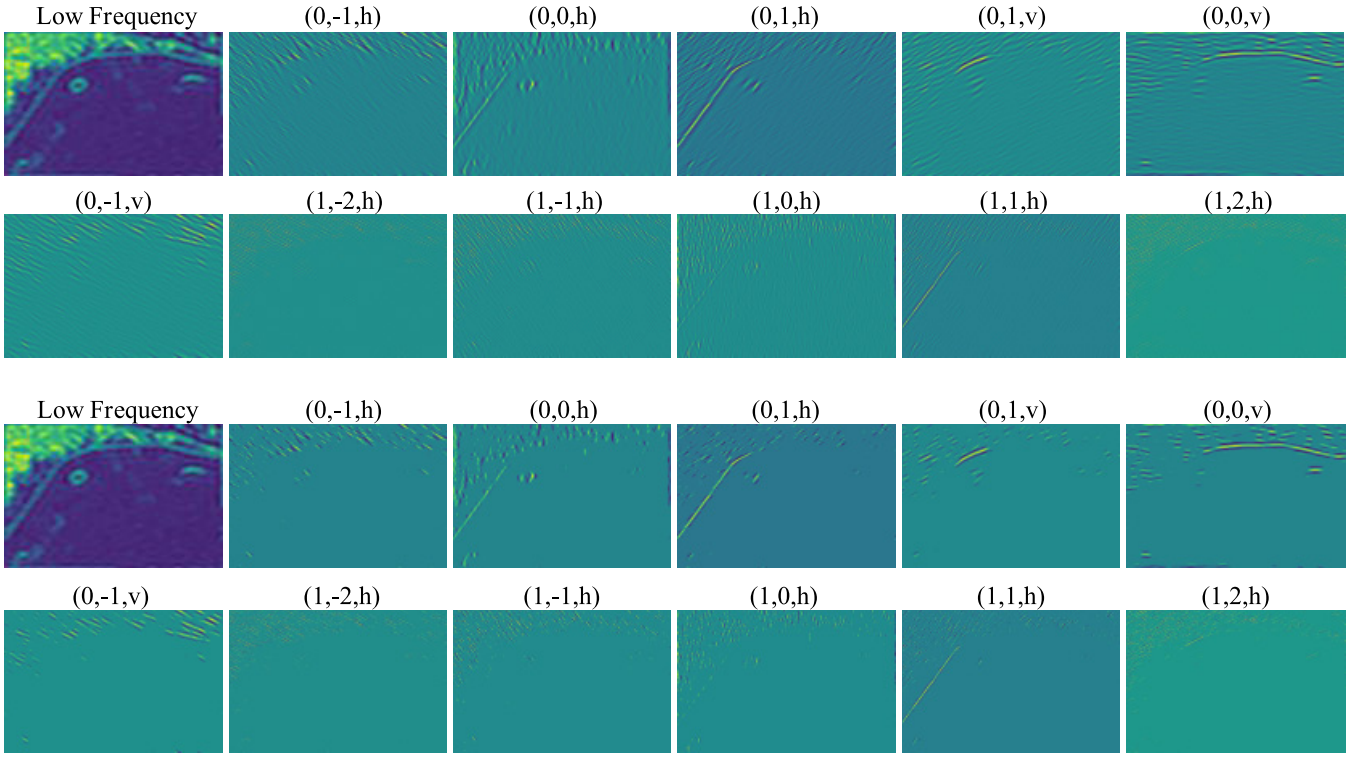


Fig. 4. 12 coefficients of the ST of the SAR image (Coastline dataset). The first two rows are the original shearlet coefficients. The last two rows are the denoised shearlet coefficients.

used as a simple trick to increase input information. Each of them first goes through the *Conv1* block with 3×3 convolution kernel. Then, the upsampling (*Upsample*) is applied to accommodate the layerwise computation of the network. After the *Upsample*, the obtained features are fed into difference SDL (DSDL).

As illustrated in Fig. 1, the DSDL is designed based on SDL. First, two branches pass through ST separately to obtain the decomposed coefficient subbands $S_1 = [S_1^L, S_1^H]$ and $S_2 = [S_2^L, S_2^H]$, both with the size of $d \times H \times W$. Then, same as the SDL, the hard threshold denoising is applied to reset the coefficient subbands and generate the new coefficient subbands $\hat{S}_1 = [S_1^L, \hat{S}_1^H]$ and $\hat{S}_2 = [S_2^L, \hat{S}_2^H]$, respectively. For the purpose of change detection, the difference level is measured in the denoised ST domain as

$$S_d = \begin{cases} \left\| \left(S_1^H\right)^2 \ominus \left(S_2^H\right)^2 \right\|_2^2 \\ \left\| \left(S_1^L\right)^2 \ominus \left(S_2^L\right)^2 \right\|_2^2 \end{cases} \tag{12}$$

where $\ominus$ denotes the subtraction operation. Compared to the usual difference calculation in the spatial domain, the differential approach in the shearlet domain is more effective in suppressing noise and capturing changed details. Finally, the differential subband coefficients are fused through addition aggregation as (11). The fused difference subband coefficients $\tilde{S}_d$ are used as the output of the DSDL.

The output of the DSDL is further fed into convolution blocks with four layers. The first three convolution blocks are followed by the *SDL*, respectively. The first convolution block uses a 5×5 filter, and the last three convolution blocks use a 3×3 filter. In the architecture, the convolution block alternates with the proposed *SDL* to learn the clean difference features and capture the key geometrical information. Following it, a fully connected layer (*FC*) is applied to classify the resulting features. The top layer is the final output with two classes of change and unchange. Table II gives the architecture of the proposed ShearNet in detail.

TABLE II
DETAILED ARCHITECTURE OF SHEARDCNETS. NOTE THAT CONV PRESENTED IN THE TABLE REFERS TO THE CONV-RELU. S AND P DENOTE STRIDE AND PADDING, RESPECTIVELY, AND $(\star, \star)$ DENOTES THE TWO BRANCHES THAT SHARE PARAMETERS

Stage	Output Size	Kernel Size	S	P
Input	$(3 \times 7 \times 7, 3 \times 7 \times 7)$			
Conv1	$(32 \times 5 \times 5, 32 \times 5 \times 5)$	$3 \times 3 \otimes 32$	1	1
Upsample	$(32 \times 16 \times 16, 32 \times 16 \times 16)$			
DSDL	$32 \times 16 \times 16$			
Conv2	$48 \times 16 \times 16$	$5 \times 5 \otimes 48$	1	2
SDL2	$48 \times 16 \times 16$			
Conv3	$96 \times 16 \times 16$	$3 \times 3 \otimes 96$	1	1
SDL3	$96 \times 16 \times 16$			
Conv4	$64 \times 16 \times 16$	$3 \times 3 \otimes 64$	1	1
SDL4	$64 \times 16 \times 16$			
Conv5	$64 \times 16 \times 16$	$3 \times 3 \otimes 64$	1	1
FC	64			
Output	2			

IV. LEARNING FROM NOISY LABELS

In this section, we show how to train ShearNet. First, noisy labels are prepared to construct the training of the network. Then, the training process is described concretely.

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A. Preparation of Noisy Labels

The ShearNet needs labeled data to fine-tune the network for learning task-specific features. In the field of SAR image change detection, the available label data are difficult to obtain. Therefore, most of the existing methods adopt the preclassification as suboptimal supervised labels, which are much cheaper and easily access. However, the existing preclassification methods are usually obtained by traditional methods, which only operate in shallow features and, thus, lack enough robustness to noise. Inspired by deep clustering, we cluster the initial deep features to generate the preclassification. Because deep-level features are more separable than shallow ones, this method can effectively improve the robustness of the result. In this deep preclassification method, the initialization mode of the network parameters and the network structure are the two main factors that affect its performance. The Kaiming initialization method is specifically designed for the ReLU activation function and proved to be superior to Xavier initialization [59]. Therefore, we use the Kaiming initialization method. Moreover, the well-designed network structure can guarantee the reliability of the preclassification results after many trials and errors.

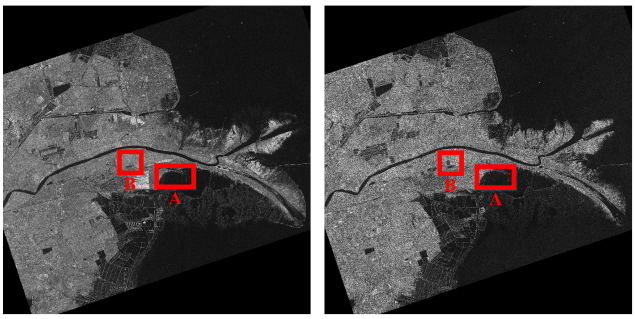
Let $X' = \{(x_{1,i}, x_{2,i})\}_{i=1}^{N}$ represent a training set, where $x_{1,i}$ and $x_{2,i}$ are the processed patches by pixelwise sliding windows from two temporal images I_1 and I_2, respectively. The initial parameters of the ShearNet are denoted as Θ. The output of the top fully connected layer thus is denoted as $F(X', \Theta)$. In a deep network, the lower layers learn the basic geometry features for the image such as edges and corners, while the higher layers capture semantic information. Therefore, the features $F(X', \Theta)$ outputted from the top fully connected layer are more distinguishable, which are used to cluster and obtain the preclassification result. Considering simplicity and effectiveness, the K-means method is specifically applied to the features $F(X', \Theta)$ and obtains the preclassification $\hat{\Omega} = \{\hat{\Omega}_u, \hat{\Omega}_c\}$, where $\hat{\Omega}_u$ denotes unchanged class and $\hat{\Omega}_c$ denotes changed class. In order to distinguish the $\hat{\Omega}_c$ class from unchanged class $\hat{\Omega}_u$, let V_{Ω_u} and V_{Ω_c} be the cluster mean features for classes $\hat{\Omega}_u$ and $\hat{\Omega}_c$, respectively. The changed area between two images should have a higher difference feature value than the unchanged area. Accordingly, the cluster that has a higher feature average value is considered as the $\hat{\Omega}_c$ class, and the other cluster is the $\hat{\Omega}_u$ class.

B. Training Procedure

With the preclassification $\hat{\Omega}$, how to obtain a more accurate detection result from the inaccurate pseudolabels is a challenging problem. Existing methods often select the pixels that are correctly classified with high probability as training samples and use the cross-entropy (CE) loss to train a classifier. In this way, not only the selection of training samples affects the quality of network learning but also the commonly used CE loss (CEL) is sensitive to noise and hinders the learning of an accurate classifier. Therefore, we introduce a novel paradigm to train the ShearNet, i.e., learning from noise labels by treating the inaccurate preclassification as noisy labels. The $\mathcal{L}_{\mathrm{DMI}}$ loss is a novel information-theoretic loss function, which is noise-robust regardless of noise pattern [60]. It can be applied to any neural network without any conditions and auxiliary costs. Therefore, $\mathcal{L}_{\mathrm{DMI}}$ is used as our loss function, which is expressed as

$$\begin{aligned}\mathcal{L}_{\mathrm{DMI}}(\Theta) &= -\log\left(\mathrm{DMI}\left(h\left(X', \Theta\right), \hat{\Omega}\right)\right) \\ &= -\log\left(\left|\det\left(Q_{h(X',\Theta),\hat{\Omega}}\right)\right|\right)\end{aligned} \tag{13}$$

where $h(X', \Theta)$ denotes the classifier with the network parameters Θ and taking the input X', $\hat{\Omega}$ is the noisy labels, i.e., the preclassification result, $Q_{h(X',\Theta),\hat{\Omega}}$ denotes the joint distribution of $h(X', \Theta)$ and $\hat{\Omega}$, and $\mathrm{DMI}(h(X', \Theta), \hat{\Omega})$ is a performance measure of the classifier $h(X', \Theta)$ based on information-theoretic. In the end, the more accurate result Ω is predicted by the trained-well networks. The use of $\mathcal{L}_{\mathrm{DMI}}$ can boost the network to break the limitation of preclassification accuracy and greatly promote the performance of the trained classifier toward being more accurate.



(a) (b)

Fig. 5. Yellow River dataset. (a) Image acquired in June 2008. (b) Image acquired in June 2009.

V. Experiments

A. Datasets and Evaluation Metrics

To show the effectiveness and robustness of the proposed method, we evaluate our method on four real SAR datasets. All of them are introduced as follows.

Yellow River Dataset: The dataset contains a pair of SAR images with the size of 7666 × 7692 pixels. It was taken by the Radarsat-2 satellite in June 2008 and June 2009, over the Yellow River Estuary. As shown in Fig. 5, it can be used to survey the changes in the region from 2008 to 2009. We select two typical sets of image pairs from the Yellow River dataset, i.e., Coastline and FarmlandC datasets, corresponding to areas A and B marked by the red box in Fig. 5. The two datasets have different ground objects with different sizes and geometries [see Fig. 6(a) and (b)].

Bern Dataset: The dataset is related to the city of Bern, Switzerland, and is acquired by the ERS-2 satellite SAR sensor in April and May 1999. It shows the area before and after the flooding [see Fig. 6(c)].

De Gaulle Airport Dataset: The dataset consists of two SAR images captured on July 24, 1997, and October 24, 1998, in the De Gaulle Airport in France [see Fig. 6(d)].

The descriptions of these datasets are summarized in Table III, including their corresponding satellites, sensor type (ST), spatial resolution (SR), and image size.

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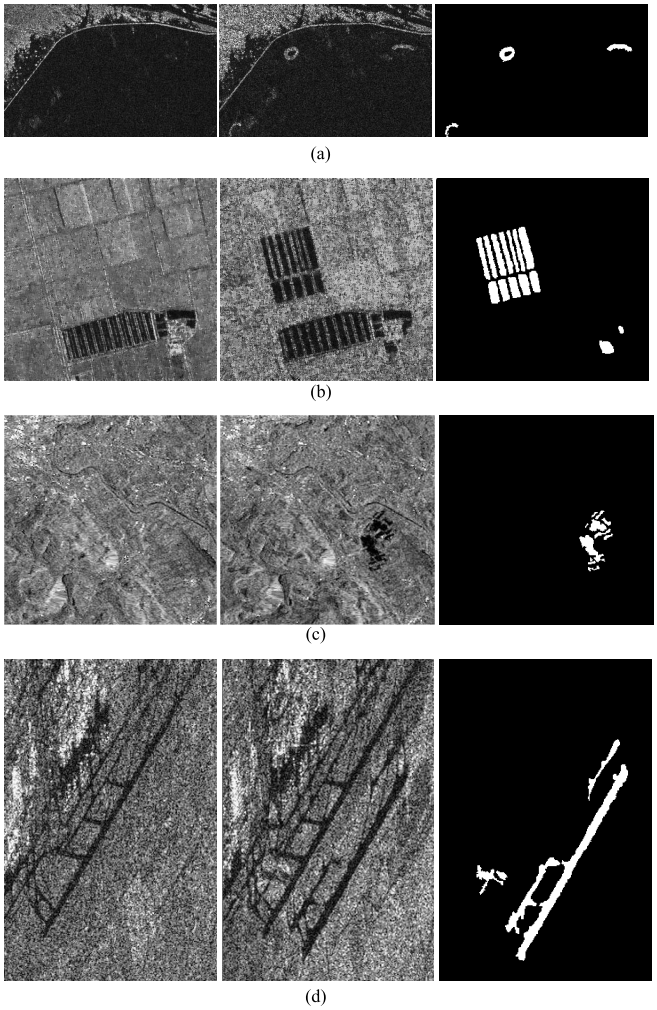


Fig. 6. Test datasets: (a) Coastline, (b) FarmlandC, (c) Bern, and (d) De Gaulle Airport.

TABLE III
DESCRIPTIONS OF USED DATASETS

Datasets	Satellite	ST	SR	Size
Coastline	Radarsat-2	SAR	8m	450×280
FarmlandC	Radarsat-2	SAR	8m	306×291
Bern	ERS-2	SAR	10m	301×301
De Gaulle Airport	ERS-1	SAR	30m	240×370

In our experiments, we use false positives (FPs), false negatives (FNs), overall error (OE), overall accuracy (OA), and the Kappa coefficient to evaluate the performance of the proposed method quantitatively. The relevant calculation formulas are shown in Table IV.

B. Implementation Details

The Adam optimizer [61] with the learning rate $\mathrm{lr} = 10^{-6}$, $\beta = (0.9, 0.999)$, and $\epsilon = 10^{-8}$ is used for training in all datasets. We adopt a minibatch size of 256. The number of epochs is usually set to 5. The network parameters Θ are initialized by the Kaiming initialization method [59]. The features $F(X', \Theta)$ are whitened to reduce the feature redundancy before clustering. Training samples are constructed by randomly sampling the same numbers N_{training} for each class, where $N_{\text{training}} = (a \times b/2) + 1$, and 2 represents the two classes of change and unchange. In all experiments, we conduct our experiments based on PyTorch 1.1.0, CUDA 10.0, and a single GeForce RTX 2070 SUPER 8-GB GPU.

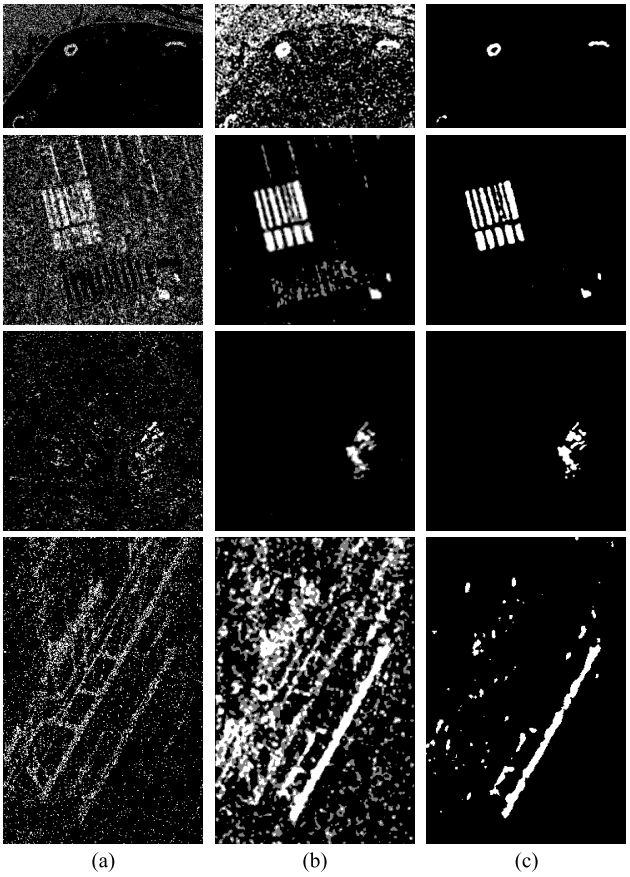


Fig. 7. Results generated by different preclassification methods: (a) JFCM, (b) HFCM, and (c) DeepCluster.

C. Detection Performance

To verify the effectiveness and robustness of the proposed method, we construct the experiments on multiple real SAR datasets.

1) Evaluation on Preclassification: In this section, two typical and related preclassification methods, JFCM [3] and HFCM [34], are introduced as the comparison methods to verify the performance of our preclassification method. The results generated by JFCM, HFCM, and DeepCluster (ours) are shown in Fig. 7. In HFCM, its result divides three classes, changed class Ω_c, unchanged class Ω_u, and indeterminate class Ω_i, in which the white part represents the changed, the black part represents the unchanged, and the while gray part indicates the areas of uncertainty. Because the pixels within Ω_c or Ω_u have a higher probability to be detected correctly, Ω_c and Ω_u are used as training samples. Therefore, the accuracy of HFCM is evaluated by Ω_c and Ω_u. The quantitative evaluations are listed in Table V.

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TABLE IV
EVALUATION CRITERIA

Evaluation Indicators	Meaning	Calculation Formula
FP	Unchanged pixels that are incorrectly detected as changed pixels	-
FN	Changed pixels that are incorrectly detected as unchanged pixels	-
OE	Overall error	$OE = FP + FN$
OA	Overall accuracy	$PCC = \frac{TP+TN}{TP+TN+FP+FN}$
Kappa	Kappa coefficient	$Kappa = \frac{PCC-PRE}{1-PRE}$ $PRE = \frac{(TP+FP)\cdot(TP+FN)+(FN+TN)\cdot(FP+TN)}{(TP+TN+FP+FN)^2}$

[1] TN (True Negative): unchanged pixels that are correctly classified. TP (True Positive): changed pixels that are correctly classified.

TABLE V
DETECTION RESULTS OF DIFFERENT PRECLASSIFICATION METHODS ON DIFFERENT DATASETS

Dataset	Method	FP(%)	FN(%)	PCC(%)	KC(%)
Coastline	JFCM [3]	4.94	0.59	94.47	13.24
	HFCM [34]	10.55	0.0008	89.44	15.05
	DeepCluster	0.02	0.26	99.71	84.72
FarmlandC	JFCM [3]	15.53	2.25	82.21	22.13
	HFCM [34]	0.02	0.24	99.74	96.24
	DeepCluster	0.04	1.45	98.50	84.83
Bern	JFCM [3]	3.81	0.82	95.36	14.58
	HFCM [34]	0.0033	0.09	99.90	91.85
	DeepCluster	0.08	0.26	99.66	85.59
De Gaulle Airport	JFCM [3]	8.32	3.36	88.32	8.80
	HFCM [34]	9.87	0.25	89.87	36.46
	DeepCluster	1.14	1.70	97.16	63.43

First, from the visual results, JFCM generates many noises on all four datasets [see Fig. 7(a)]. This is because the FCM algorithm only considers the shallow gray-level information, which is susceptible to speckle noise. HFCM implements the FCM algorithm in a coarse-to-fine procedure; thus, it performs better than JFCM, especially on FarmlandC and Bern datasets. However, HFCM fails to detect changed areas better for Coastline and De Gaulle Airport datasets [see Fig. 7(b)]. Because deep-level features are more discriminating, the DeepCluster algorithm is more robust than JFCM and HFCM, and it achieves better performance on all datasets. Quantitative evaluation in Table V also shows the HFCM algorithm achieves the best accuracy on FarmlandC and Bern datasets, but it is worse far from DeepCluster for other datasets [see Fig. 7(c)]. Benefitting from the multilayer structure of CNN, the DeepCluster algorithm reduces most noises and has the best robustness among these algorithms.

2) Proposed Versus State-of-the-Art Change Detection Algorithms: To demonstrate the advanced performance of the proposed method, six detectors are evaluated for comparison. They are GKI [17], PCA-Kmeans (PCAKM) [13], NRCR [62], GaborPCANet [39], CWNN [34], and WaveNet [37]. Under the same network structure as ShearNet, WaveNet adopts DWT_{ll} proposed in [37] to replace the SDL to filter noise and remain geometric structure information.

Tables VI–IX list the quantitative evaluations on different datasets. Fig. 8 shows their visual results. GKI adopts the Kittler and Illingworth minimum-error thresholding algorithm that models an SAR-specific parametric distribution for ratio image. Due to the diversity and complexity of the image distribution, the best threshold value is difficult to acquire. As shown in Fig. 8(a), GKI yields many white noisy

TABLE VI
DETECTION RESULTS OF DIFFERENT METHODS ON COASTLINE DATASET

Methods	FP(%)	FN(%)	OE(%)	PCC(%)	Kappa(%)
GKI [17]	15.30	0.21	15.52	84.48	8.05
PCAKM [13]	32.73	0.0	32.73	67.27	4.15
NRCR [62]	24.07	0.01	24.08	75.91	6.17
GaborPCANet [39]	12.61	0.01	12.62	87.38	12.70
CWNN [34]	0.82	0.07	0.89	99.11	68.55
WaveNet [37]	0.18	0.07	0.25	99.75	88.80
ShearNet (ours)	0.08	0.12	0.20	99.80	90.51

TABLE VII
DETECTION RESULTS OF DIFFERENT METHODS ON FARMLANDC DATASET

Methods	FP(%)	FN(%)	OE(%)	PCC(%)	Kappa(%)
GKI [17]	0.14	2.33	2.47	97.52	73.11
PCAKM [13]	1.46	0.55	2.00	97.99	83.21
NRCR [62]	2.07	0.76	2.83	97.17	76.98
GaborPCANet [39]	0.06	1.46	1.52	98.48	84.65
CWNN [34]	0.53	0.70	1.23	98.77	88.80
WaveNet [37]	0.48	0.57	1.05	98.95	90.53
ShearNet (ours)	0.33	0.68	1.01	99.00	90.73

TABLE VIII
DETECTION RESULTS OF DIFFERENT METHODS ON BERN DATASET

Methods	FP(%)	FN(%)	OE(%)	PCC(%)	Kappa(%)
GKI [17]	0.48	0.09	0.57	99.42	80.34
PCAKM [13]	0.27	0.13	0.40	99.60	84.78
NRCR [62]	0.64	0.10	0.74	99.26	75.87
GaborPCANet [39]	0.04	0.48	0.52	99.48	75.23
CWNN [34]	0.09	0.25	0.34	99.66	85.56
WaveNet [37]	0.26	0.12	0.38	99.62	85.56
ShearNet (ours)	0.18	0.14	0.32	99.68	87.41

TABLE IX
DETECTION RESULTS OF DIFFERENT METHODS ON DE GAULLE AIRPORT DATASET

Methods	FP(%)	FN(%)	OE(%)	PCC(%)	Kappa(%)
GKI [17]	0.26	2.89	3.15	96.85	38.50
PCAKM [13]	24.46	0.42	24.88	75.12	16.23
NRCR [62]	24.74	0.47	25.21	74.79	15.72
GaborPCANet [39]	13.97	0.41	14.39	85.61	28.21
CWNN [34]	2.27	1.25	3.52	96.48	58.53
WaveNet [37]	3.39	1.11	4.50	95.50	56.55
ShearNet (ours)	1.60	1.43	3.03	96.97	64.03

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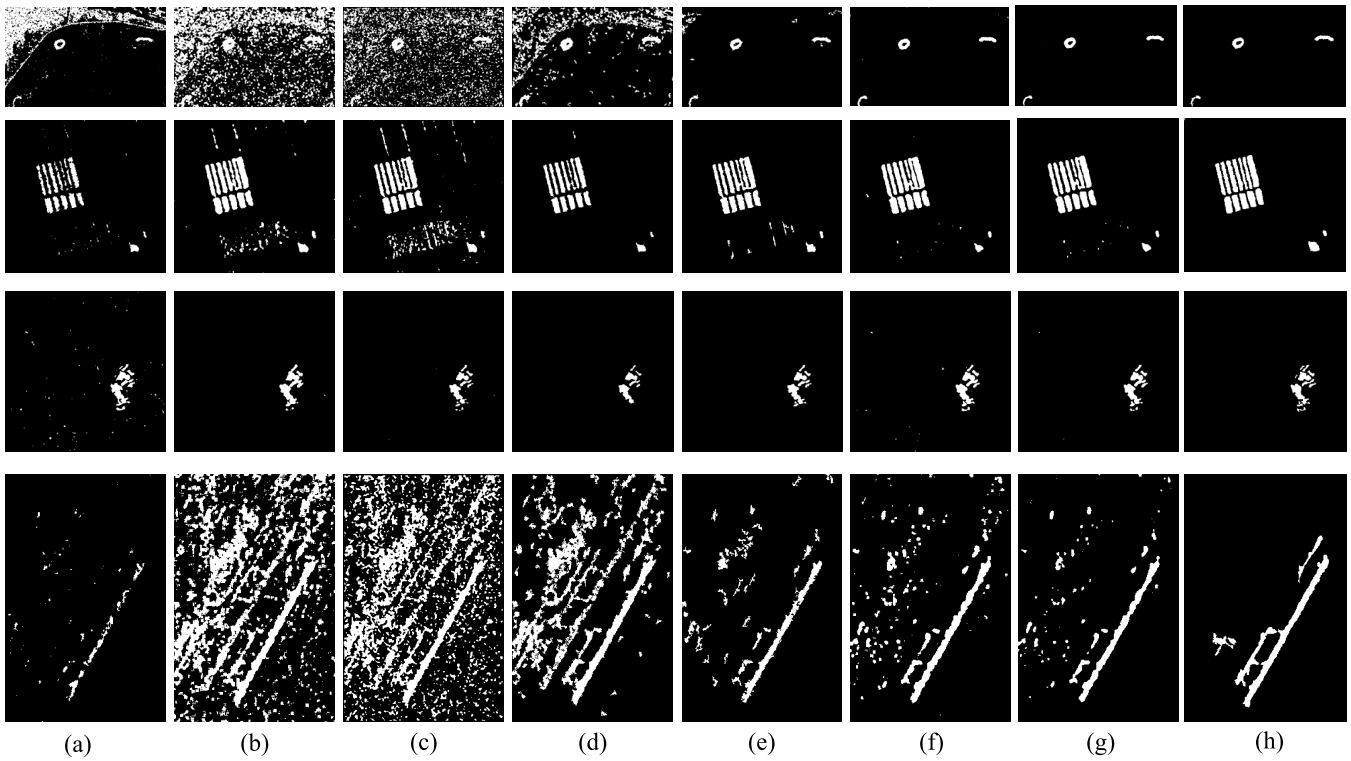


Fig. 8. Results generated by different detection methods: (a) GKI, (b) PCAKM, (c) NRCR, (d) GaborPCANet, (e) CWNN, (f) WaveNet, (g) ShearNet, and (h) ground truth.

regions on Coastline and Bern datasets; for other datasets, it misses a lot of changed details. PCAKM and NRCR both are traditional machine learning methods that use handcraft features, which lacks the adaptability for different distributions of images. Thus, they achieve similar detection results, where most unchanged pixels are recognized as noise [see Fig. 8(b) and (c)]. GaborPCANet and CWNN both are based on deep networks and, thus, perform better than GKI, PCAKM, and NRCR. However, there are still many noises especially for Coastline and De Gaulle Airport datasets [see Fig. 8(d) and (e)]. This is because GaborPCANet and CWNN both use traditional methods to obtain preclassification, which limits the ability of network learning. ShearNet improves the preclassification with the DeepCluster method and enhances the representation ability of common CNN with SDL. Compared to Wavelet transform, ST has a greater advantage in maintaining the information of edge structure. Therefore, ShearNet outperforms WaveNet and achieves the best performance among compared methods. Quantitative evaluations in Tables VI–IX also show that the performance of ShearNet is the best in suppressing noises and detecting the changed details.

D. Ablation Study

This section shows the ablation experiments to validate the effectiveness of each component in the proposed method.

1) Ablation of SDL: To verify the effectiveness of the designed SDL for SAR image change detection, we constructed multiple contrast experiments. For convenience, we have the following notation.

CNN: The common CNN with the proposed architecture listed in Table II, in which SDL is not introduced. It is considered our baseline.

ShearNet-LL: All SDLs use the low-frequency subband as output.

ShearNet-T: In SDL, all high-frequency subbands use the same threshold T that is computed by taking the median of all high-frequency subbands.

ShearNet-Ti: The median value is computed for each subband in high frequency. Then, each subband performs the threshold function separately.

ShearNet-Ti (w/o DSDL): Compared to ShearNet-Ti (with DSDL), DSDL is removed, and its difference level is computed in the spatial domain. ShearNet-LL and ShearNet-T are the variants of ShearNet-Ti (with DSDL). They calculate the difference in the ST domain. The difference between them is the threshold calculation way. As shown in Table X, ShearNet-T and ShearNet-Ti both outperform ShearNet-LL on four datasets. As we stated earlier, only using low-frequency information easily causes the loss of useful details contained in high frequency and, thus, degrades the performance of the pixelwise task. Furthermore, each subband has a different direction; using the same threshold on all subband coefficients also easily causes the inefficiency of noise reduction and the loss of useful information. Thus, ShearNet-Ti performs better than ShearNet-T in general. Common CNN and ShearNet with different schemes of SDL perform similarly on the first

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TABLE X
COMPARISON WITH DIFFERENT WAYS OF SDL FOR DENOISING

Dataset	Method	FP(%)	FN(%)	PCC(%)	KC(%)
Coastline	CNN	0.07	0.12	99.80	90.53
	ShearNet-LL	0.08	0.12	99.79	90.13
	ShearNet-T	0.09	0.11	99.80	90.63
	ShearNet-Ti(w/o DSDL)	0.08	0.11	99.80	90.66
	ShearNet-Ti(with DSDL)	0.08	0.12	99.80	90.54
FarmlandC	CNN	0.28	0.69	99.03	91.01
	ShearNet-LL	0.50	0.53	98.96	90.64
	ShearNet-T	0.32	0.65	99.02	91.01
	ShearNet-Ti(w/o DSDL)	0.30	0.65	99.04	91.16
	ShearNet-Ti(with DSDL)	0.28	0.72	99.00	90.74
Bern	CNN	0.20	0.13	99.67	87.40
	ShearNet-LL	0.21	0.13	99.66	87.01
	ShearNet-T	0.19	0.15	99.66	86.87
	ShearNet-Ti(w/o DSDL)	0.15	0.17	99.68	87.20
	ShearNet-Ti(with DSDL)	0.18	0.14	99.68	87.41
De Gaulle	CNN	0.20	1.33	96.47	61.14
	ShearNet-LL	2.13	1.32	96.54	61.67
	ShearNet-T	1.86	1.36	96.78	63.15
	ShearNet-Ti(w/o DSDL)	1.98	1.35	96.66	62.26
	ShearNet-Ti(with DSDL)	1.60	1.43	96.97	64.03

TABLE XI
COMPARISON WITH DIFFERENT LAYERS INSERTED SDL ON DE GAULLE AIRPORT DATASET

Layer-1	Layer-2	Layer-3	Layer-4	Layer-5	Kappa(%)
✓					60.25
✓	✓				62.37
✓	✓	✓			62.80
✓	✓	✓	✓		64.03
✓	✓	✓	✓	✓	63.44

three datasets, but ShearNet is obviously better than CNN on the De Gaulle Airport dataset. There are similar contrast results between ShearNet-Ti (w/o DSDL) and ShearNet-Ti (with DSDL). In sum, ShearNet-Ti (with DSDL) has better robustness in detecting changes for different scenarios of SAR data. Thus, ShearNet-Ti (with DSDL) is adopted and denoted as ShearNet for simplicity. As shown in Fig. 6(d), the De Gaulle Airport dataset has a richer texture structure that ST is good at capturing. This also suggests that ShearNet has a wider application, especially for data with rich geometric texture structures.

2) Ablation of Network Architecture with SDL: To enhance the recognition ability of common CNN, SDL is gradually embedded from the first layer (Layer-1) to the fifth layer (Layer-5). As shown in Table XI, when SDL is introduced in different layers, the network behaves differently on the De Gaulle Airport dataset. The network performs best when SDL is introduced in the first four layers. In our experiments, the setting with the best result is adopted.

3) Ablation of $\mathcal{L}_{\text{DMI}}$ Loss: We introduce $\mathcal{L}_{\text{DMI}}$ loss (DMIL), expecting to reduce the effect of noisy labels on network training. Fig. 9 shows the performance of ShearNet with different loss functions. It can be observed that the network with DMIL achieves better accuracy than that of CEL on the first three datasets. On the De Gaulle Airport dataset, the network with different loss functions produces similar

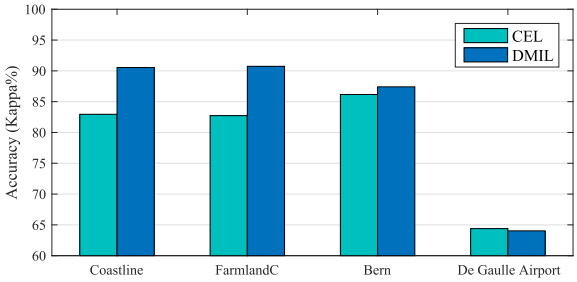


Fig. 9. Effect of different loss functions.

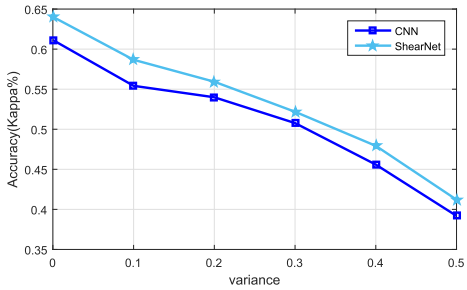


Fig. 10. Effect of different levels of noise.

detection results. As we stated earlier, DMIL is noise-robust, which is a better choice for training the network with inexact label data. It is worth noting that CEL is used to first pretrain a classifier, and then, DMIL is used as a refined fine-tuning to improve the performance of the classifier in the original paper [60]. Here, we extended its applicability. We only use the DMIL function to train the network during the whole training, simplifying the training process.

E. Effect of Noise

Since our method integrates a denoising algorithm, it is necessary to analyze the denoising effect on change detection under different noise conditions. Therefore, we applied different levels of speckle noise to the input image (De Gaulle Airport dataset). The resulting performance is shown in Fig. 10. As the noise variance increases (mean = 0), the performance of both CNN and ShearNet decreases, but ShearNet is always better than CNN. This indicates that ShearNet is more robust to noise than basic CNN.

F. Denoising as Preprocessing

In SAR image change detection, denoising is usually preprocessing to reduce the effect of speckle noise. To this end, we test the performance of common CNN when the shearlet-based hard threshold approach (CNN + preprocess) shown in Fig. 2 is applied as preprocessing. Table XII lists the compared results. It can be observed that CNN, CNN + preprocess, and ShearNet produce similar performance on Coastline and FarmlandC datasets. On other datasets, CNN and ShearNet both outperform CNN + preprocess. Especially for the De Gaulle Airport dataset, ShearNet achieves far better performance than the other two methods. These results suggest that CNN + preprocess can achieve slight gains on

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TABLE XII
COMPARISON BETWEEN DENOISING OPERATION AS PREPROCESSING AND DENOISING IN CNN

Dataset	Method	FP(%)	FN(%)	PCC(%)	KC(%)
Coastline	CNN	0.07	0.12	99.80	90.53
	CNN+preprocess	0.07	0.12	99.81	90.88
	ShearNet	0.08	0.12	99.80	90.54
FarmlandC	CNN	0.28	0.69	99.03	91.01
	CNN+preprocess	0.32	0.62	99.06	91.41
	ShearNet	0.28	0.72	99.00	90.74
Bern	CNN	0.20	0.13	99.67	87.40
	CNN+preprocess	0.15	0.18	99.67	86.78
	ShearNet	0.18	0.14	99.68	87.41
De Gaulle Airport	CNN	0.20	1.33	96.47	61.14
	CNN+preprocess	2.60	1.57	95.83	54.74
	ShearNet	1.60	1.43	96.97	64.03

some datasets. However, the proposed ShearNet that integrates shearlet-based denoising into CNN is more robust and powerful for change detection, because it performs very well in suppressing noise and maintaining changed details for more datasets with different geometric features.

VI. CONCLUSION

In this article, we present a deep ShearNet for SAR image change detection by introducing SDL. In the shearlet domain, the characteristics of signal and noise show that the noise is usually isolated points with small amplitude, while the valid signal is usually continuous areas with large amplitude. Therefore, the SDL first transforms the CNN feature into the shearlet domain and then combines it with the threshold method to filter noise. Benefiting from the optimal sparse representation of ST, the ShearNet that integrates SDL is able to better preserve the geometric structures of the image while reducing noise during network inference. Moreover, to avoid the use of manually annotated data, we use DeepCluster to obtain preclassification results as pseudolabels and introduce noise-robust loss to reduce the effect of mislabels on training. Experimental results on four SAR datasets show that ShearNet provides more availability and is more robust to the complex geometry of SAR images. In future work, we will try to combine local and global information to improve the effectiveness of the detection of changed regions with CNN. We also want to try the idea on Mindspore.

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